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CU24250178 roll no:60
labsheet -2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
print("liberaries are installed")
liberaries are installed
df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-
selectedcolumns.csv") print(df.head())
```

```
   Age Attrition     BusinessTravel  DailyRate              Department

    41       Yes      Travel_Rarely       1102                   Sales

\
0

1   49        No  Travel_Frequently        279  Research & Development

2   37       Yes      Travel_Rarely       1373  Research & Development

3   33        No  Travel_Frequently       1392  Research & Development

4   27        No      Travel_Rarely        591  Research & Development


   DistanceFromHome  Education EducationField  EmployeeCount
EmployeeNumbe
                  1          2  Life Sciences              1 0
                                                             1
                  8          1  Life Sciences              1 1
                                                             2
                  2          2          Other              1 2
                                                             4
                  3          4  Life Sciences              1 3
                                                             5
                  2          1        Medical              1 4
                                                             7
```

```
print("Shape of Dataset:", df.shape)

print("\nColumn Names:")
print(df.columns)
```

```
print("\nData Types:")
print(df.dtypes)
```

```
print("\nDataset Information:")
df.info()

print("\nSummary Statistics:")
print(df.describe(include="all"))
```

```
Shape of Dataset: (1470, 10)

Column Names:
Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate',
'Department',
       'DistanceFromHome', 'Education', 'EducationField',
'EmployeeCount',
       'EmployeeNumber'],
dtype='str')

Data Types:
Age                 int64
Attrition             str
BusinessTravel        str
DailyRate           int64
Department            str
DistanceFromHome    int64
Education           int64
EducationField        str
EmployeeCount       int64
EmployeeNumber      int64
dtype: object

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
0   Age               1470 non-null   int64
1   Attrition         1470 non-null   str
2   BusinessTravel    1470 non-null   str
3   DailyRate         1470 non-null   int64
4   Department        1470 non-null   str
5   DistanceFromHome  1470 non-null   int64
6   Education         1470 non-null   int64
7   EducationField    1470 non-null   str
```

```
8   EmployeeCount     1470 non-null   int64  9   EmployeeNumber
    1470 non-null   int64
dtypes: int64(6), str(4)
memory usage: 115.0 KB

Summary Statistics:
                Age Attrition BusinessTravel     DailyRate  \ count
1470.000000     1470          1470  1470.000000   unique         NaN
2                3           NaN    top                NaN         No
Travel_Rarely         NaN   freq           NaN     1233         1043
NaN   mean      36.923810       NaN            NaN   802.485714   std
9.135373       NaN            NaN   403.509100   min       18.000000
NaN              NaN    102.000000    25%        30.000000        NaN
NaN   465.000000
50%       36.000000       NaN            NaN   802.000000   75%
43.000000       NaN            NaN  1157.000000   max
60.000000       NaN            NaN  1499.000000

                    Department  DistanceFromHome    Education
EducationField  \
count                     1470       1470.000000  1470.000000
1470
unique                       3               NaN          NaN
6
top     Research & Development               NaN          NaN  Life
Sciences
freq                       961               NaN          NaN
606
mean                       NaN          9.192517     2.912925
NaN
std                        NaN          8.106864     1.024165
NaN
min                        NaN          1.000000     1.000000
NaN
25%                        NaN          2.000000     2.000000
NaN
50%                        NaN          7.000000     3.000000
NaN
75%                        NaN         14.000000     4.000000
NaN
max                        NaN         29.000000     5.000000
NaN

        EmployeeCount  EmployeeNumber
count          1470.0     1470.000000
unique            NaN             NaN  top
NaN             NaN  freq             NaN
NaN  mean               1.0     1024.865306
std               0.0      602.024335  min
```

```
1.0        1.000000  25%               1.0
491.250000
50%               1.0     1020.500000  75%
1.0     1555.750000  max               1.0
2068.000000
```

```python
for column in df.columns:
    print("\n", column)
    print(df[column].value_counts().head(10))
```

```
 Age
Age
35    78
34    77
36    69
31    69
29    68
32    61
30    60
33    58
38    58
40    57
Name: count, dtype: int64

 Attrition
Attrition
No     1233
Yes     237
Name: count, dtype: int64

 BusinessTravel
BusinessTravel
Travel_Rarely        1043
Travel_Frequently     277
Non-Travel            150
Name: count, dtype: int64

 DailyRate
DailyRate
691     6
408     5
1082    5
530     5
1329    5
329     5
334     4
1125    4
427     4
1225    4
Name: count, dtype: int64

 Department
Department
Research & Development    961
Sales                     446
```

```
Human Resources             63
Name: count, dtype: int64

 DistanceFromHome
DistanceFromHome
2     211
1     208
10     86
9      85
3      84
7      84
8      80
5      65
4      64
6      59
Name: count, dtype: int64

 Education
Education
3    572
4    398
2    282
1    170
5     48
Name: count, dtype: int64

 EducationField
EducationField
Life Sciences       606
Medical             464
Marketing           159
Technical Degree    132
Other                82
Human Resources      27
Name: count, dtype: int64

 EmployeeCount
EmployeeCount
1    1470
Name: count, dtype: int64

 EmployeeNumber
EmployeeNumber
1     1
2     1
4     1
5     1
7     1
8     1
10    1
```

```
11    1
12    1
13    1
Name: count, dtype: int64
```

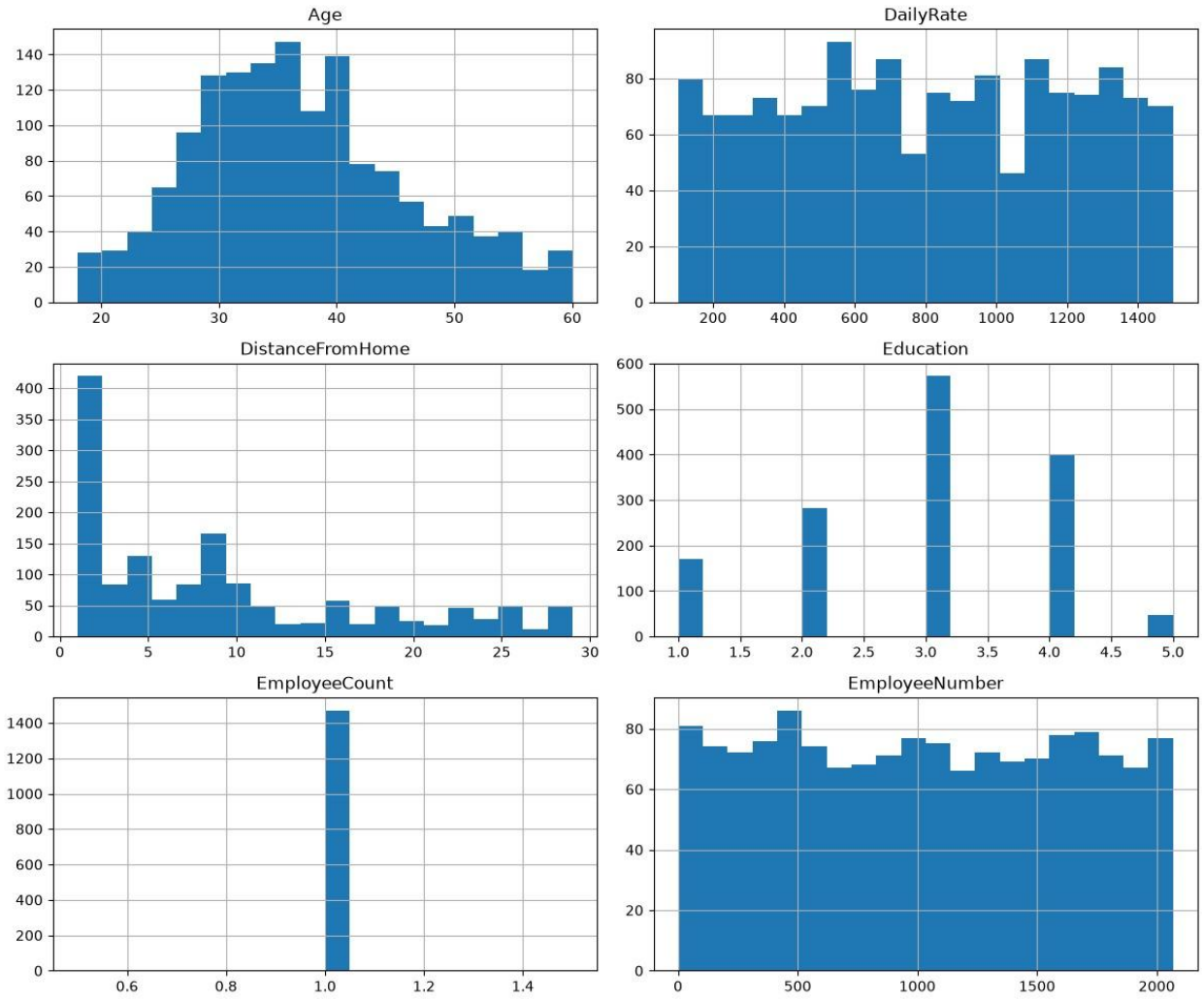
```
numerical_columns = df.select_dtypes(include=np.number).columns

print("Numerical Columns:")
print(numerical_columns)
```

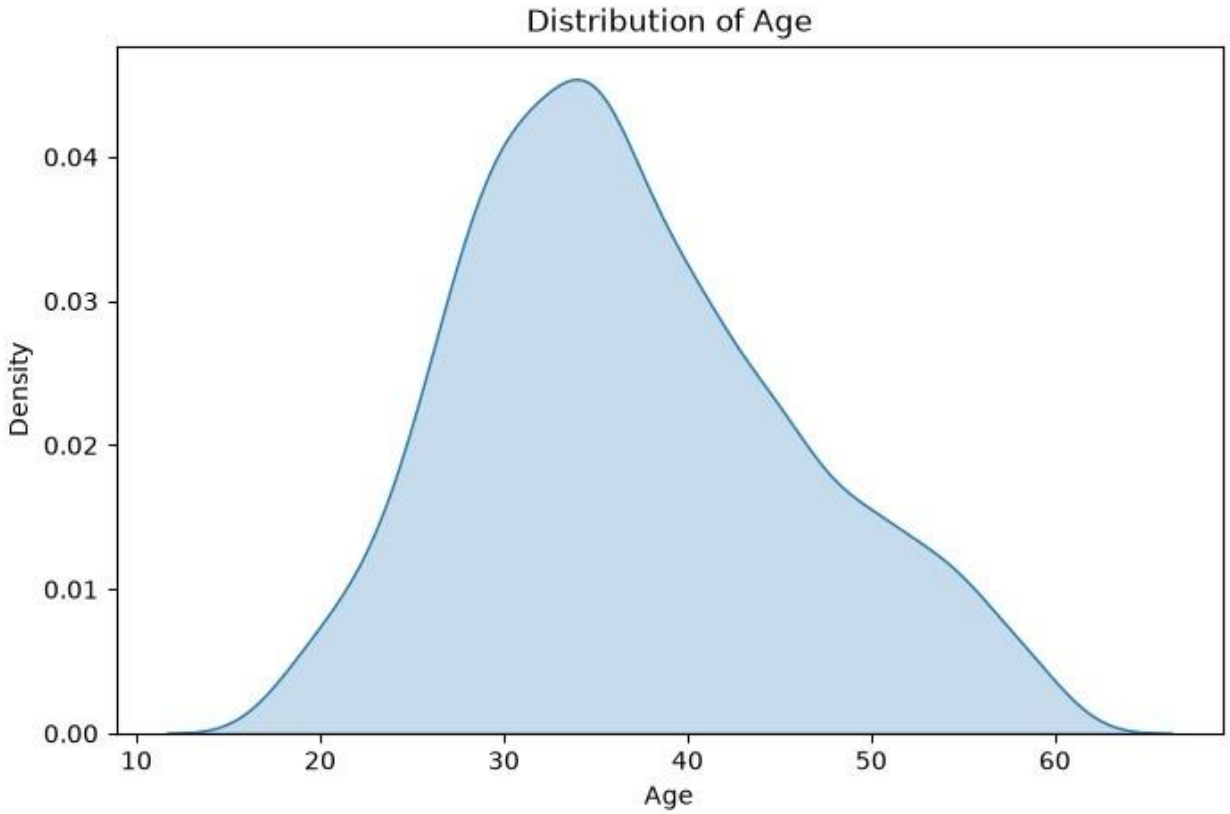
```
Numerical Columns:
Index(['Age', 'DailyRate', 'DistanceFromHome', 'Education',
'EmployeeCount',
       'EmployeeNumber'],
      dtype='str')
```

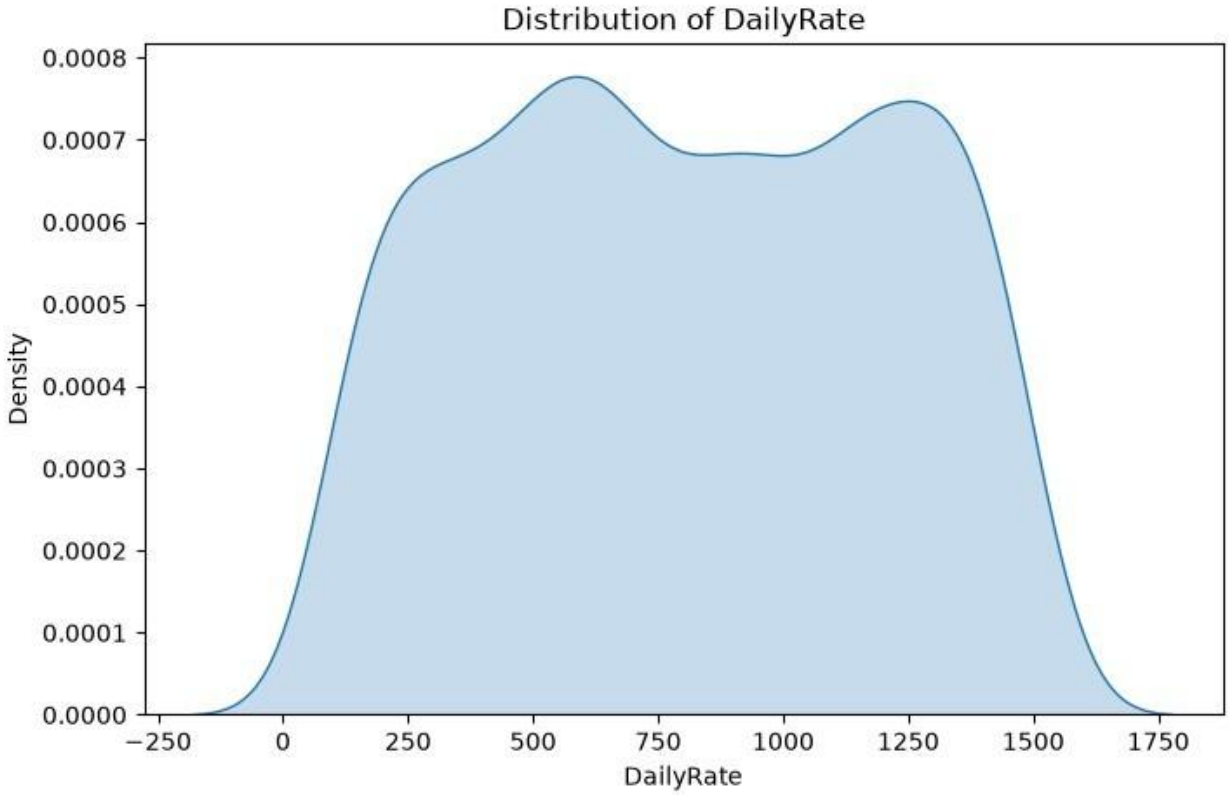
```
df[numerical_columns].hist(
    figsize=(12, 10),
    bins=20
)

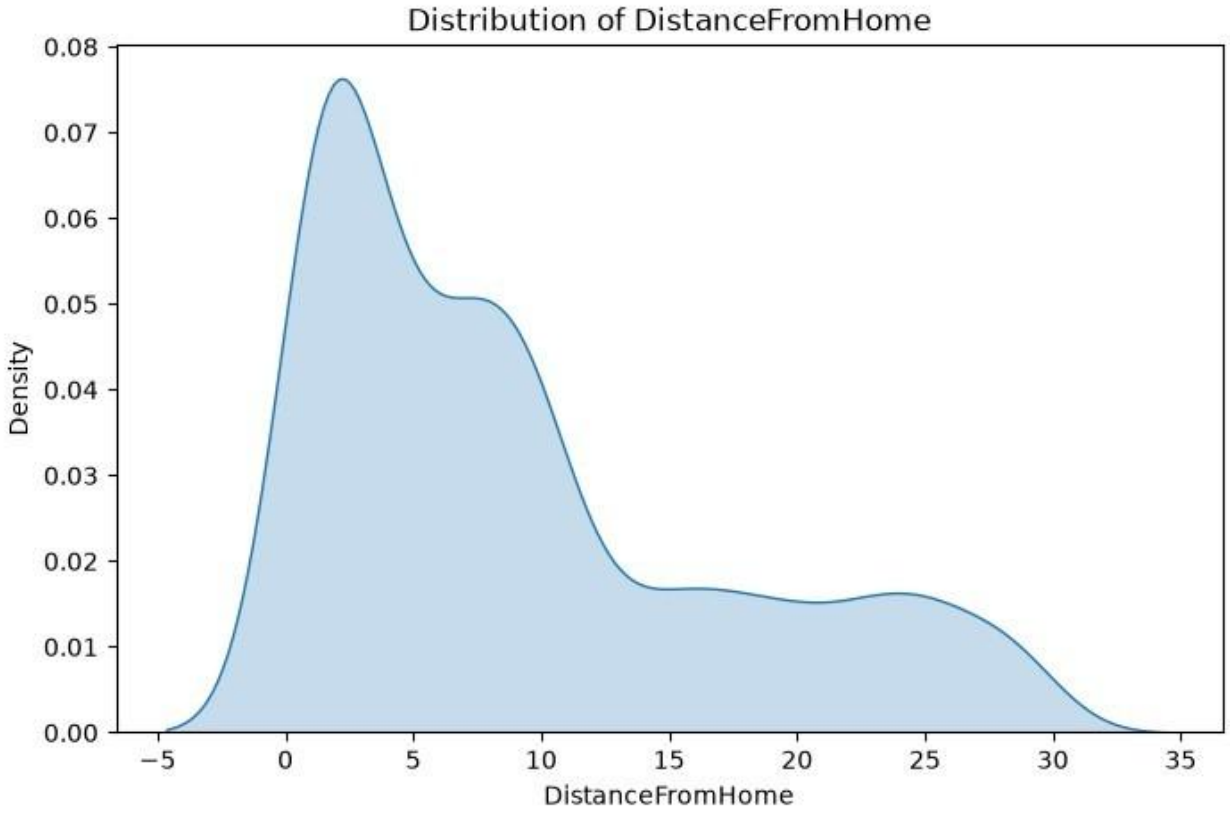
plt.tight_layout()
plt.show()
```

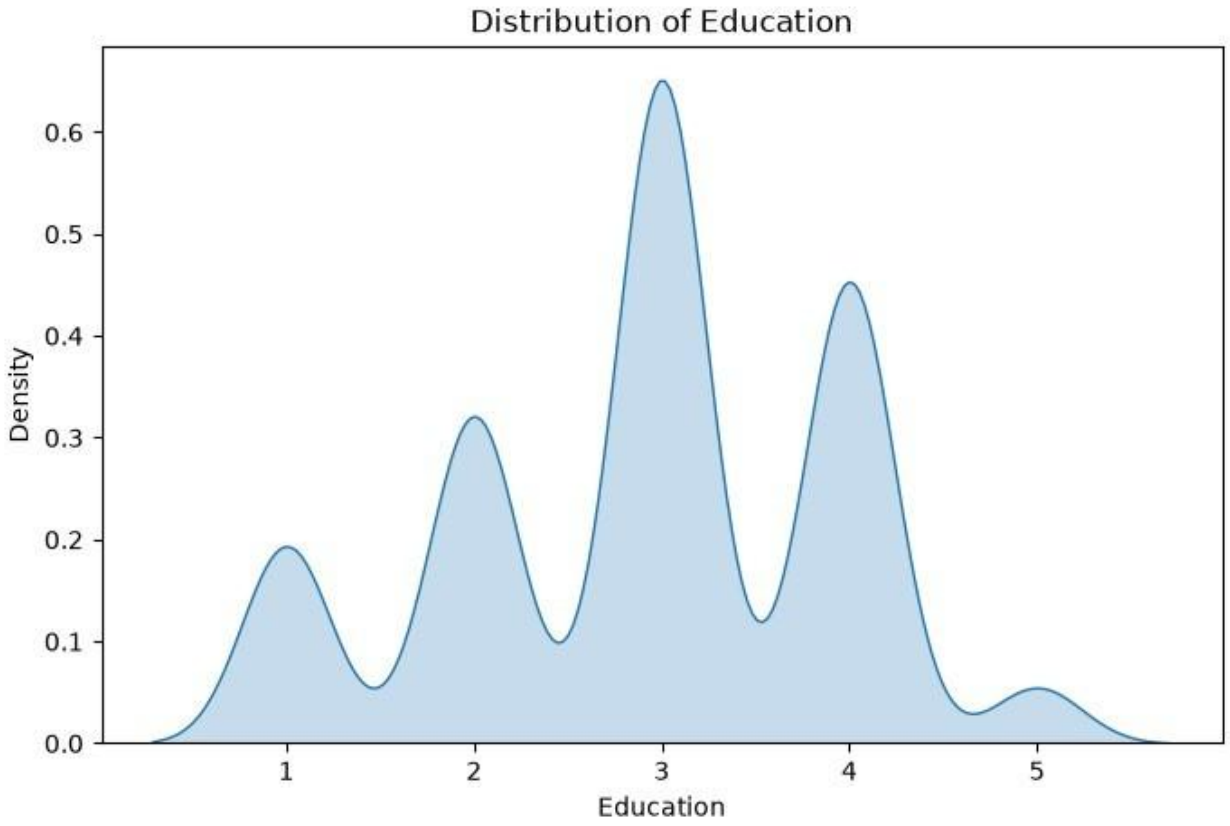



```
for column in numerical_columns:
plt.figure(figsize=(8, 5))
sns.kdeplot(data=df, x=column, fill=True)
plt.title(f"Distribution of {column}")
plt.xlabel(column)     plt.ylabel("Density")
plt.show()
```

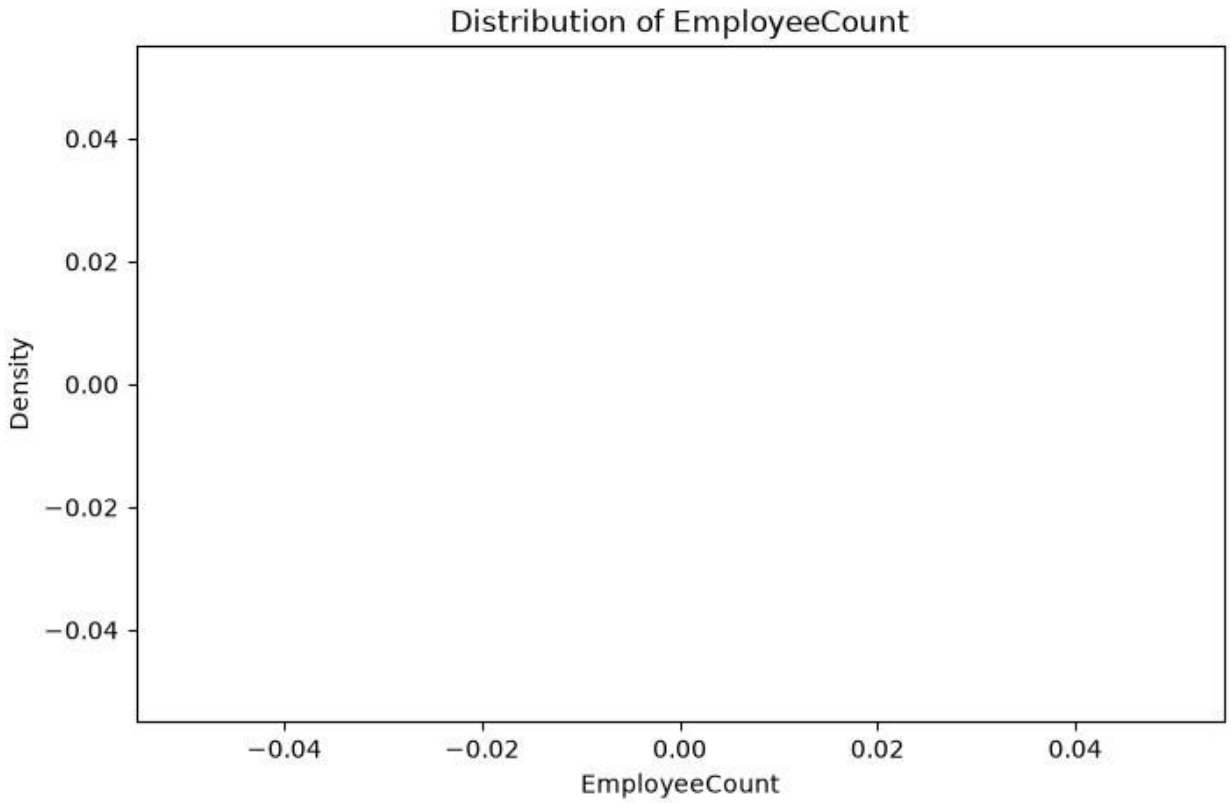

Distribution of Age
Density
0.00
0.01
0.02
0.03
0.04
10
20
30
40
50
60
Age

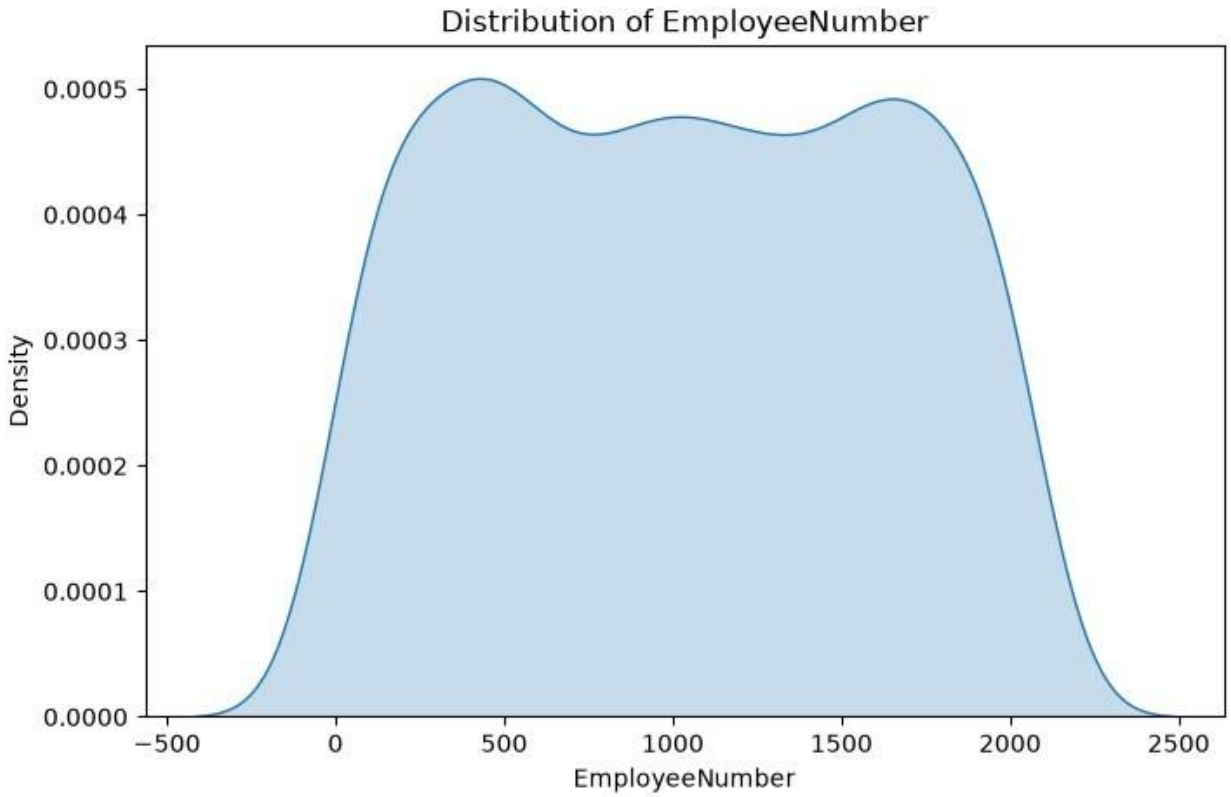

Distribution of DailyRate
Density
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
0.0006
0.0007
0.0008
−250
0
250
500
750
1000
1250
1500
1750
DailyRate


Distribution of DistanceFromHome
Density
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
−5
0
5
10
15
20
25
30
35
DistanceFromHome


Distribution of Education
Density
0.6
0.5
0.4
0.3
0.2
0.1
0.0
1
2
3
4
5
Education

```
/tmp/ipykernel_36896/2863598134.py:3: UserWarning: Dataset has 0
variance; skipping density estimate. Pass `warn_singular=False` to
disable this warning.
  sns.kdeplot(data=df, x=column, fill=True)
```






```
categorical_columns = df.select_dtypes(
    include=["object", "category"]
).columns

print("Categorical Columns:")
print(categorical_columns)
```

```
Categorical Columns:
Index(['Attrition', 'BusinessTravel', 'Department', 'EducationField'],
dtype='str')
```

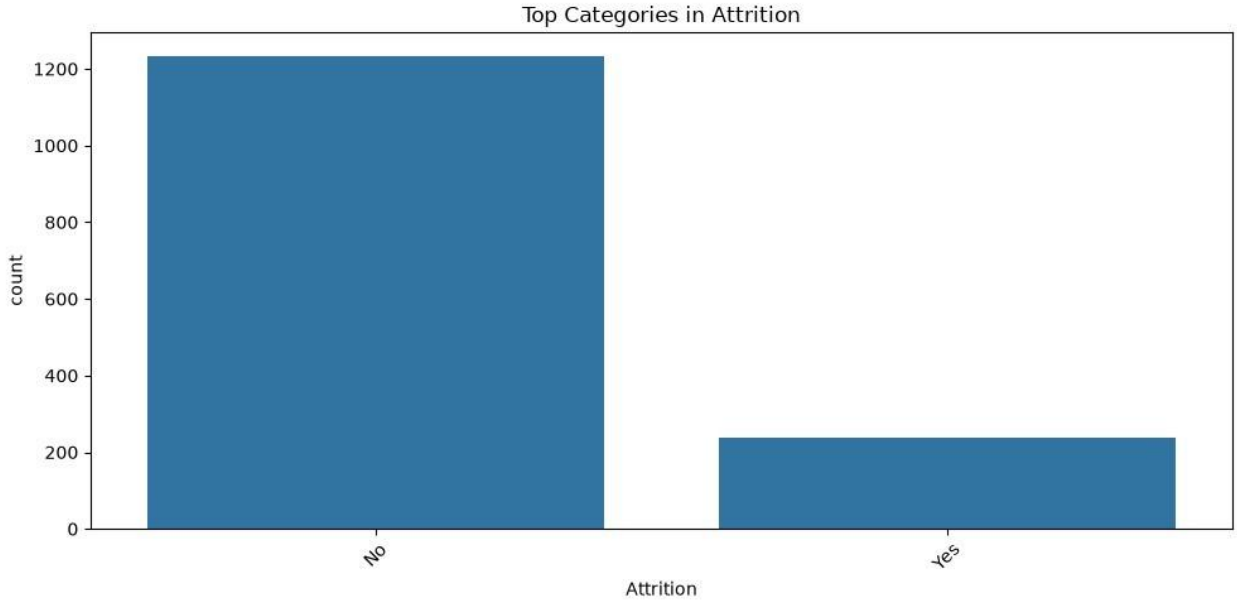
```
/tmp/ipykernel_36896/1130369359.py:1: Pandas4Warning: For backward
compatibility, 'str' dtypes are included by select_dtypes when
'object' dtype is specified. This behavior is deprecated and will be
removed in a future version. Explicitly pass 'str' to `include` to
select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-
strings.html#string-migration-select-dtypes for details on how to
write code that works with pandas 2 and 3.
  categorical_columns = df.select_dtypes(
```

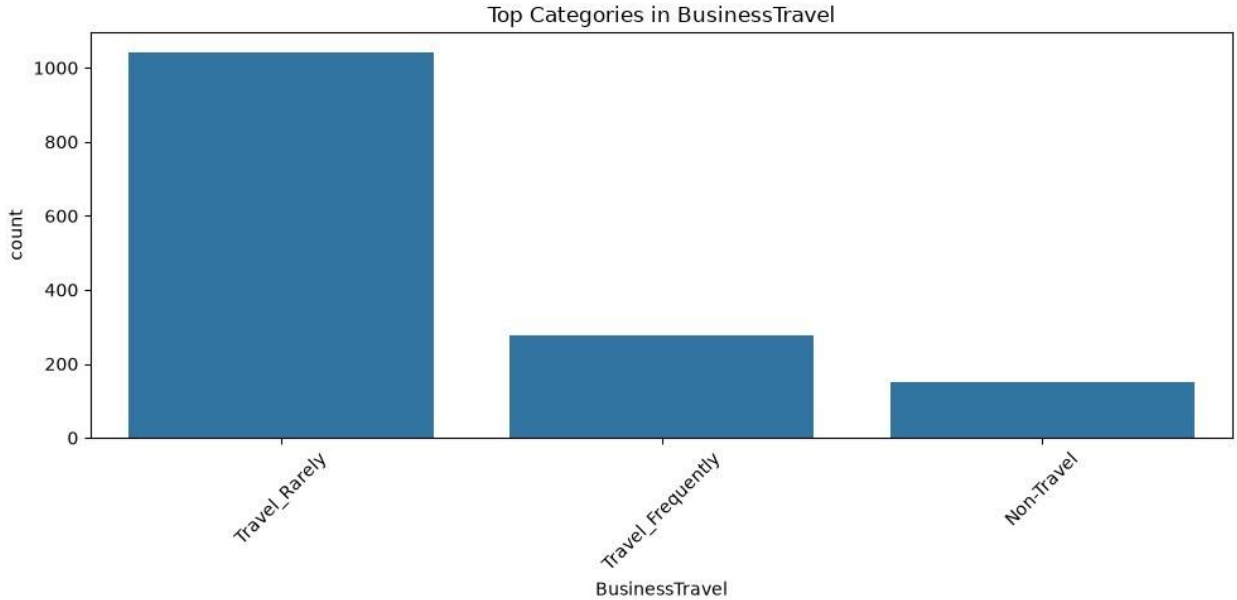
```
for column in categorical_columns:
    plt.figure(figsize=(10, 5))

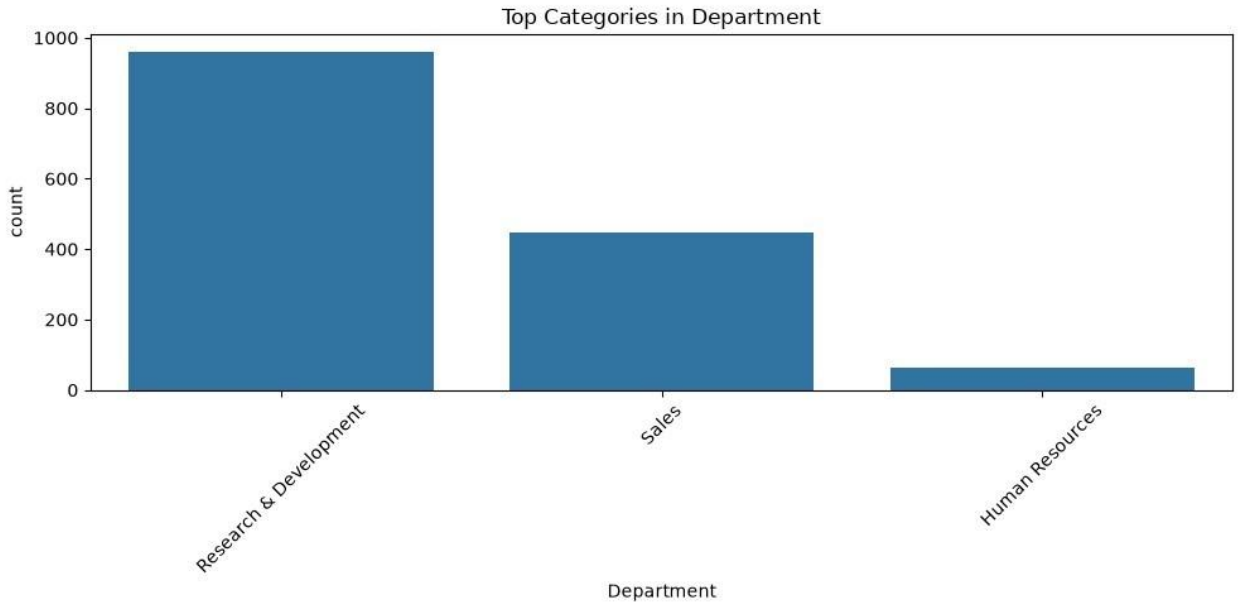
    sns.countplot(
```

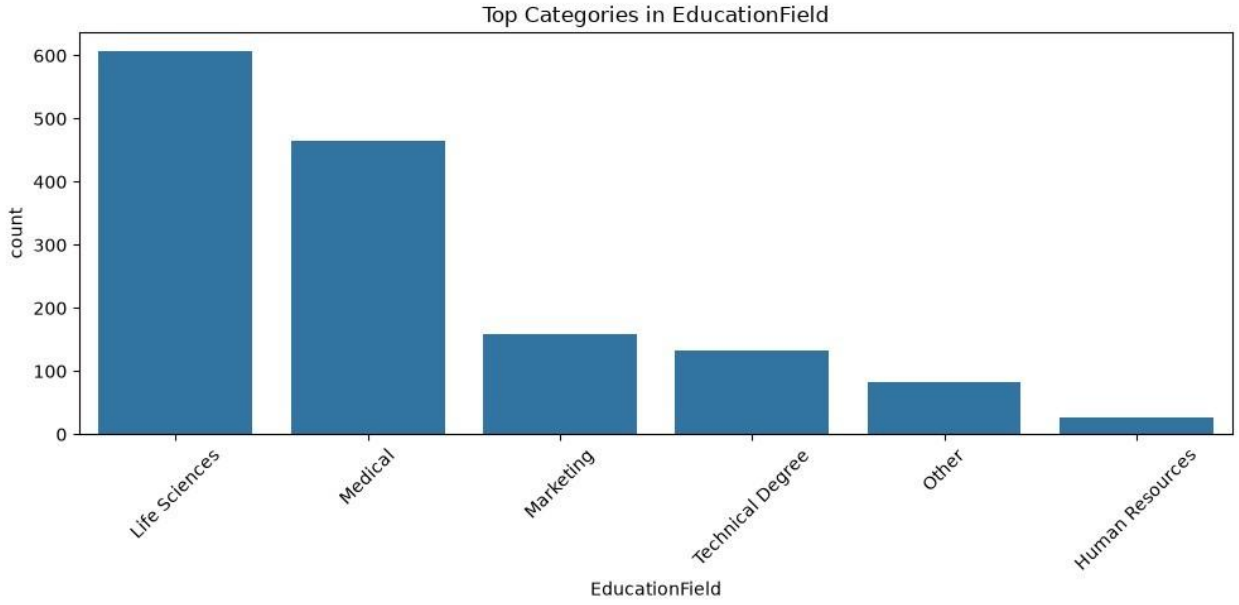
```
        data=df,
x=column,
        order=df[column].value_counts().head(10).index
    )

    plt.title(f"Top Categories in {column}")
    plt.xticks(rotation=45)
plt.tight_layout()
    plt.show()
```










```
correlation = df[numerical_columns].corr()
print(correlation)
```

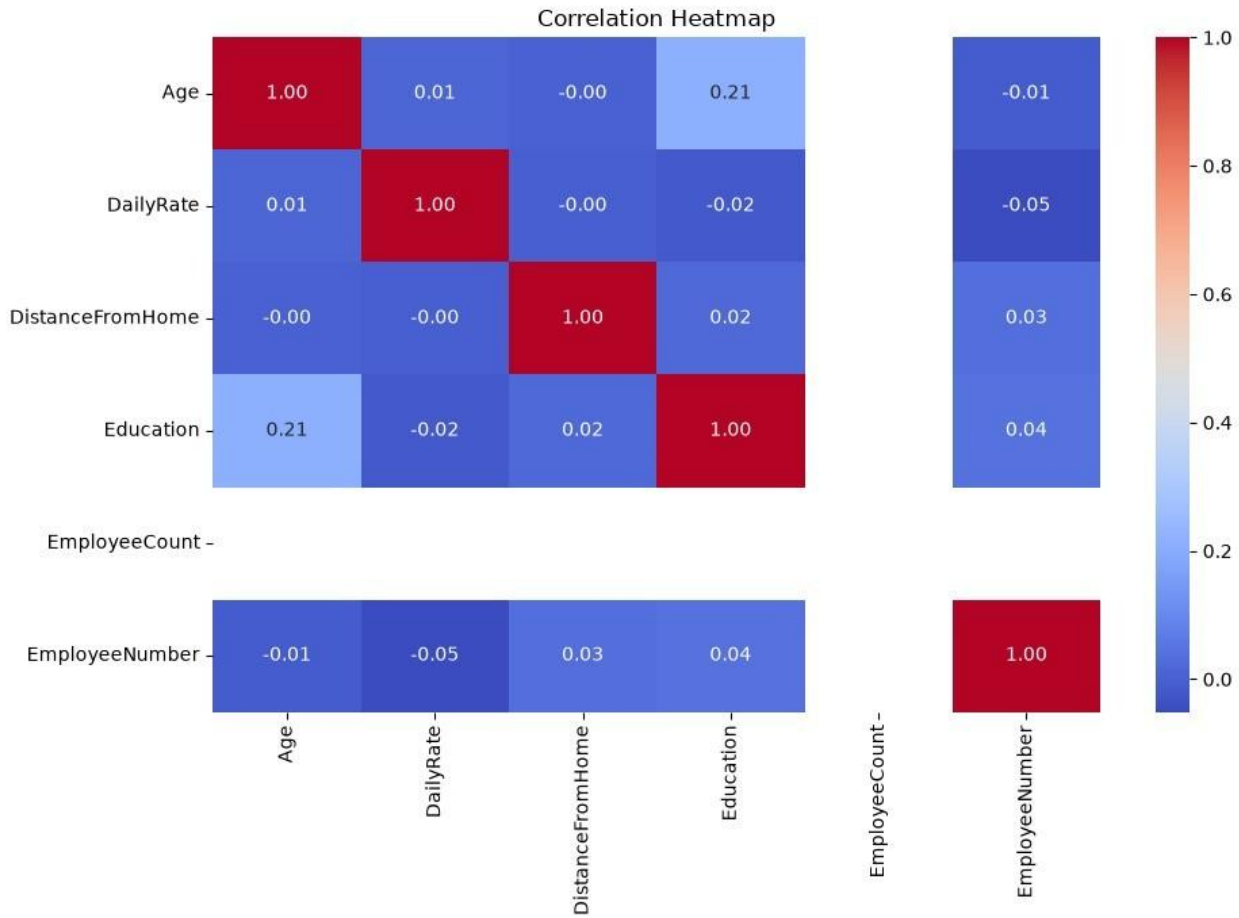
```
                          Age  DailyRate  DistanceFromHome  Education  \
Age                  1.000000   0.010661         -0.001686   0.208034
DailyRate            0.010661   1.000000         -0.004985  -0.016806
DistanceFromHome    -0.001686  -0.004985          1.000000   0.021042
Education            0.208034  -0.016806          0.021042   1.000000
EmployeeCount             NaN        NaN               NaN        NaN
EmployeeNumber      -0.010145  -0.050990          0.032916   0.042070
```

```
                    EmployeeCount  EmployeeNumber
Age                           NaN       -0.010145
DailyRate                     NaN       -0.050990
DistanceFromHome              NaN        0.032916
Education                     NaN        0.042070
EmployeeCount                 NaN             NaN
EmployeeNumber                NaN        1.000000
```

```
plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

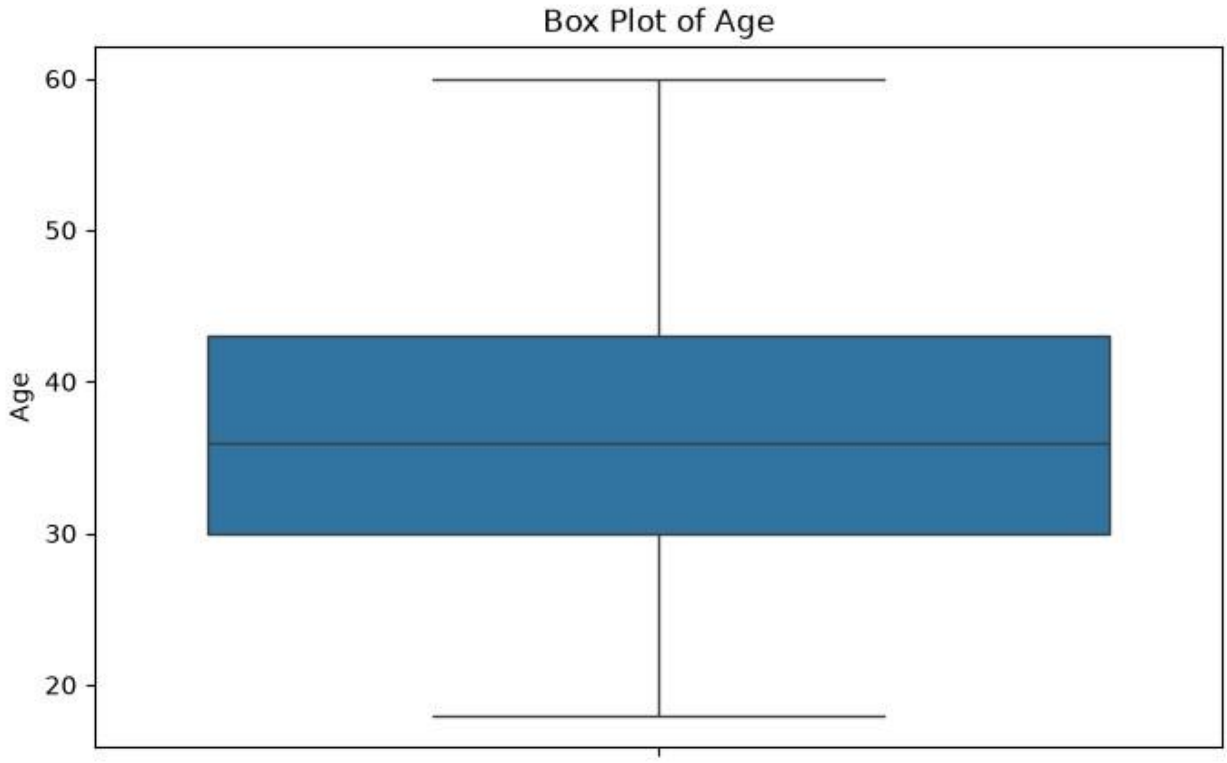
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

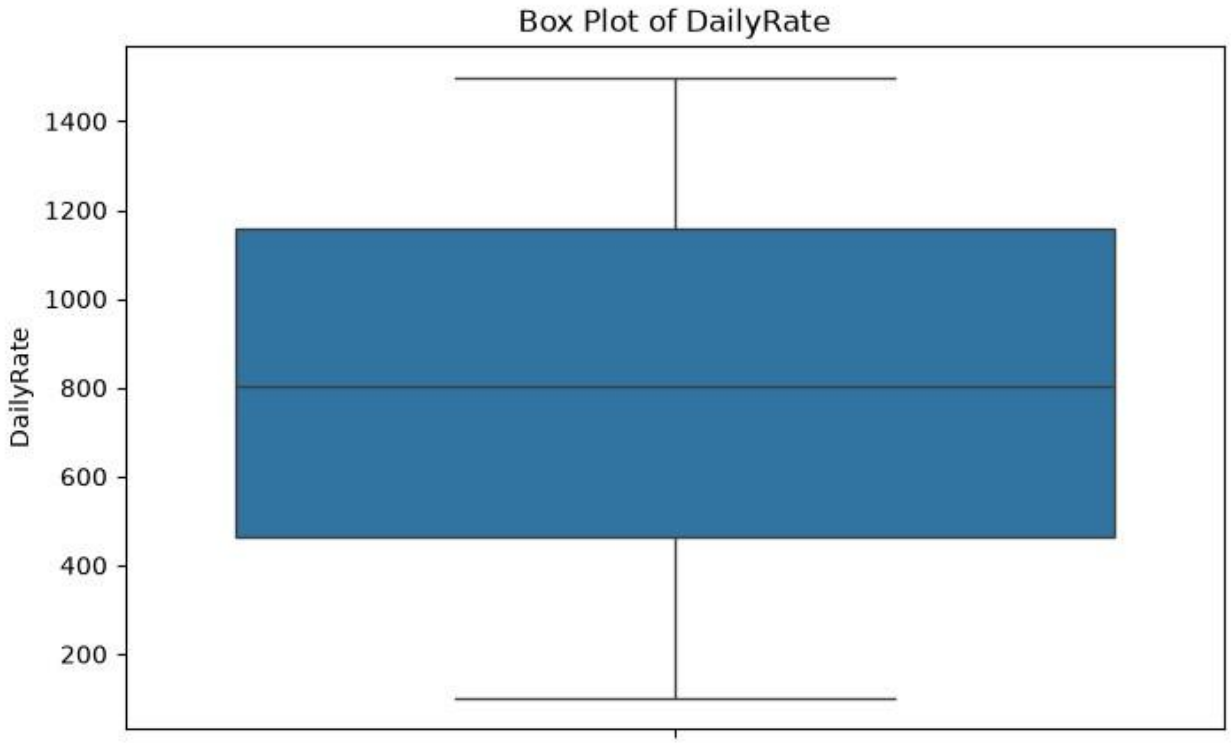


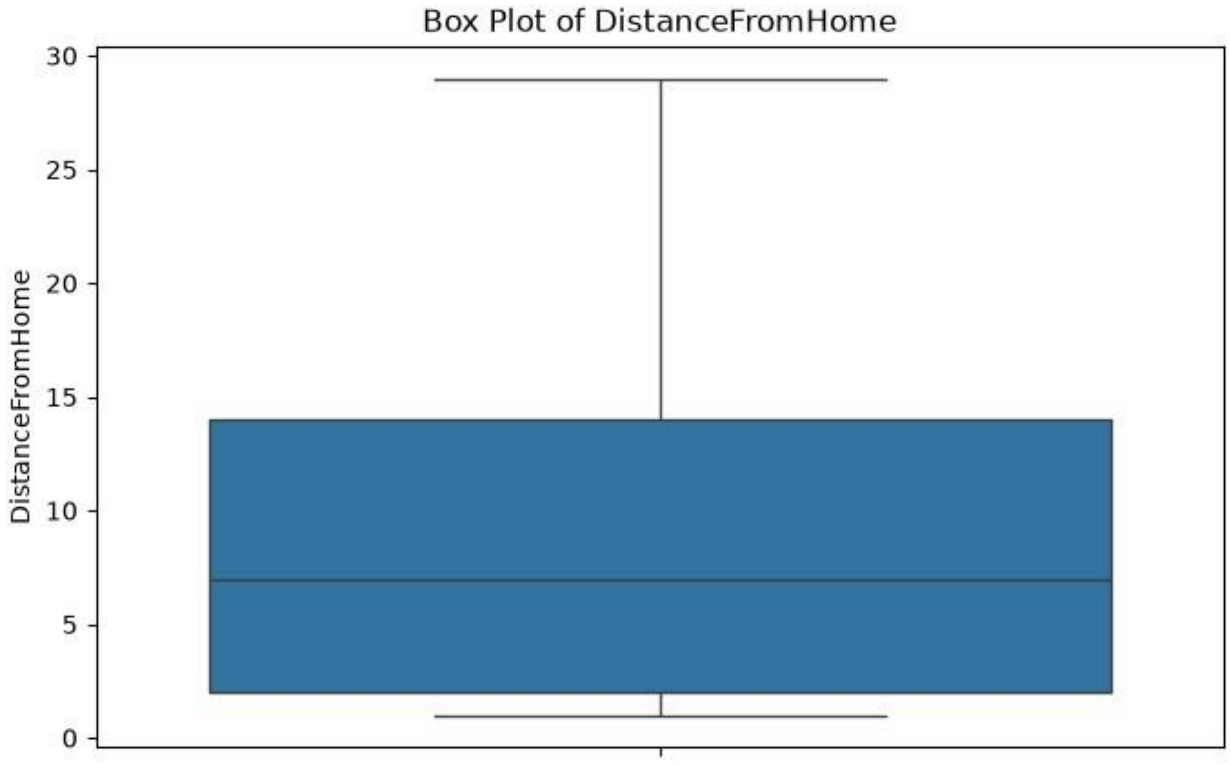
```
for column in numerical_columns:
plt.figure(figsize=(8, 5))

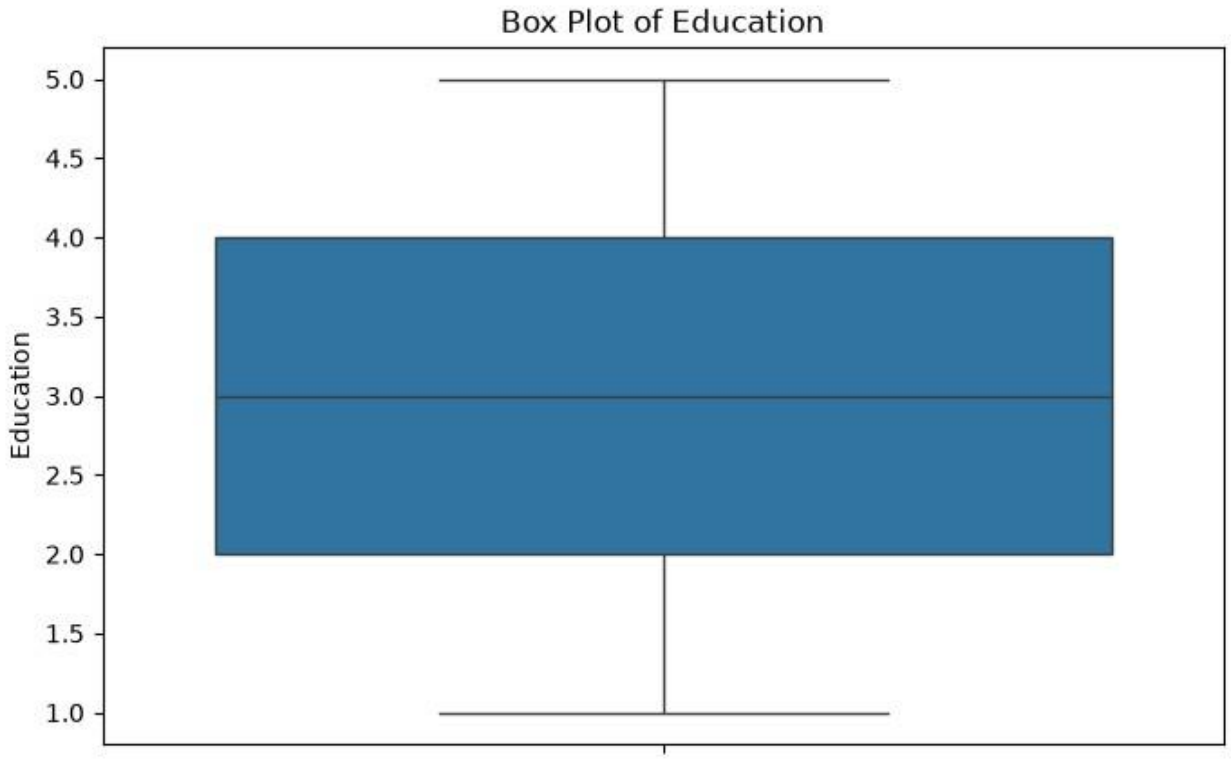
    sns.boxplot(data=df, y=column)

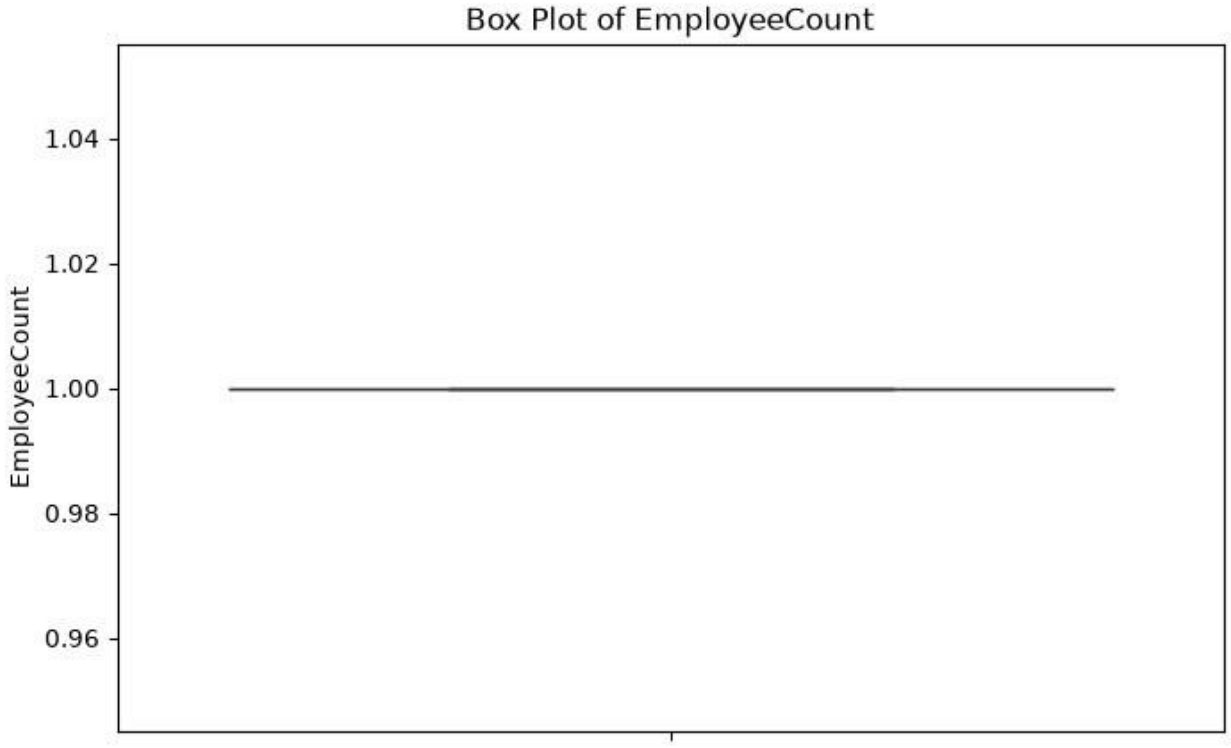
    plt.title(f"Box Plot of {column}")
plt.show()
```

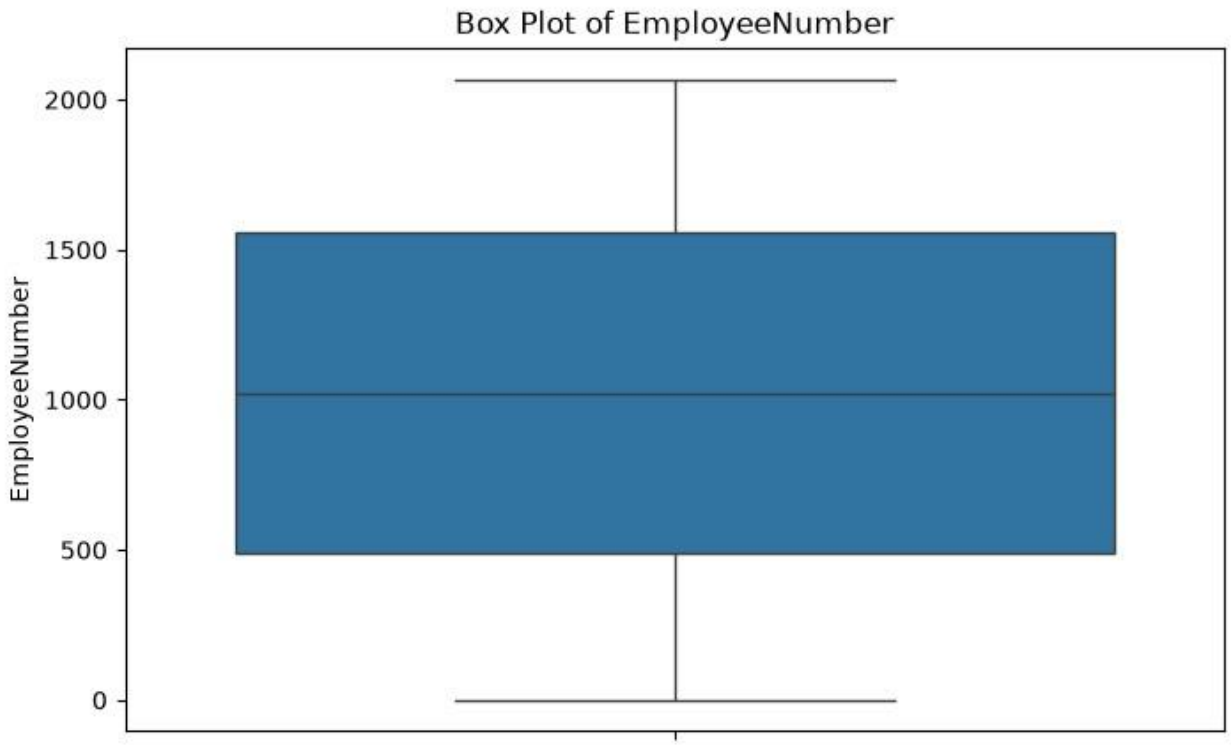



Box Plot of DailyRate
DailyRate
1400
1200
1000
800
600
400
200


Box Plot of DistanceFromHome
DistanceFromHome
30
25
20
15
10
5
0


Box Plot of Education
Education
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0


Box Plot of EmployeeCount
EmployeeCount
1.04
1.02
1.00
0.98
0.96



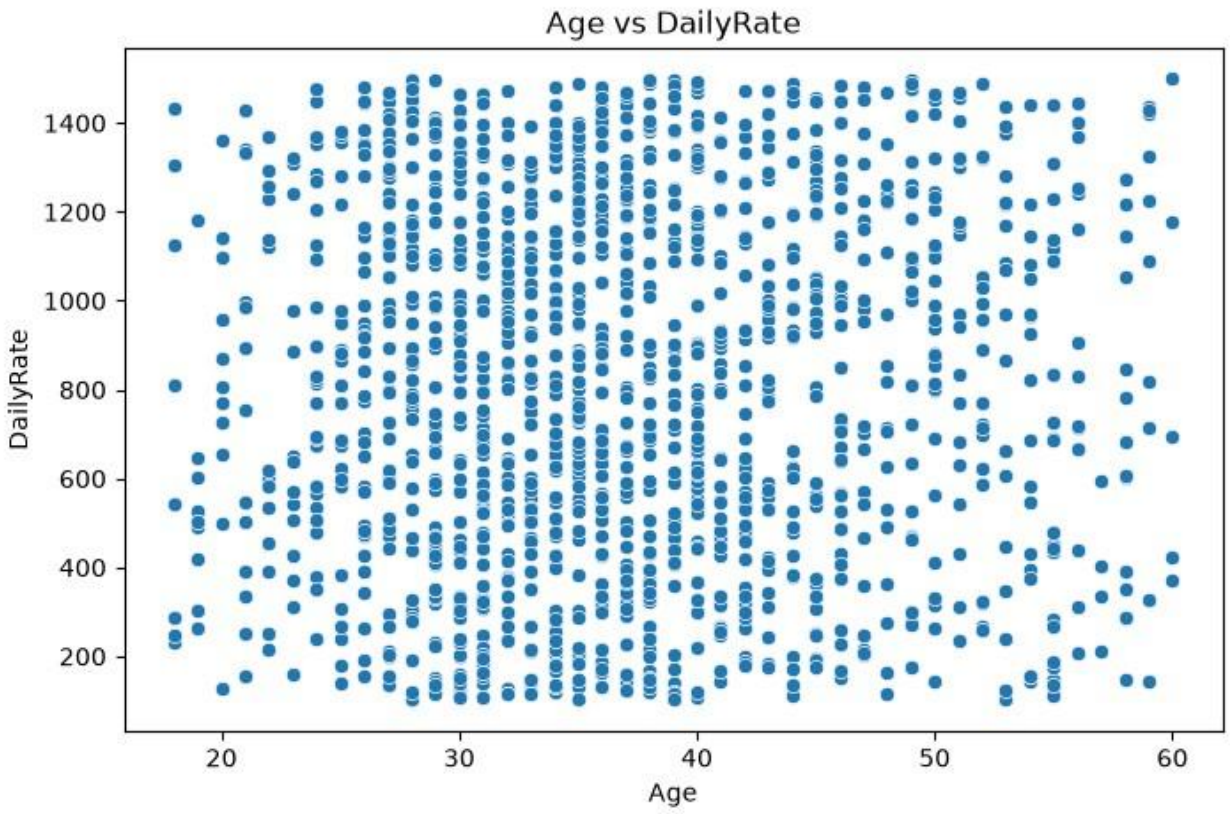

```
if len(numerical_columns) >= 2:

    x = numerical_columns[0]
y = numerical_columns[1]

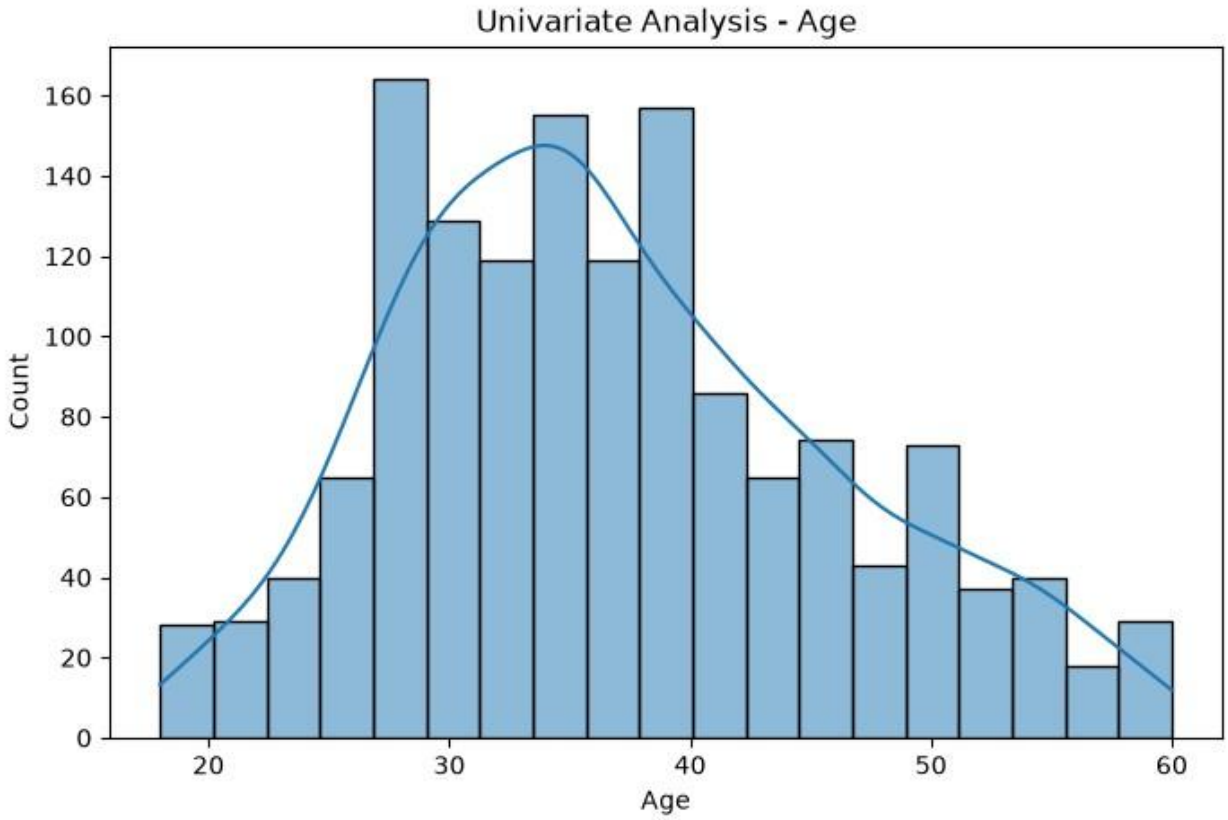
    plt.figure(figsize=(8, 5))

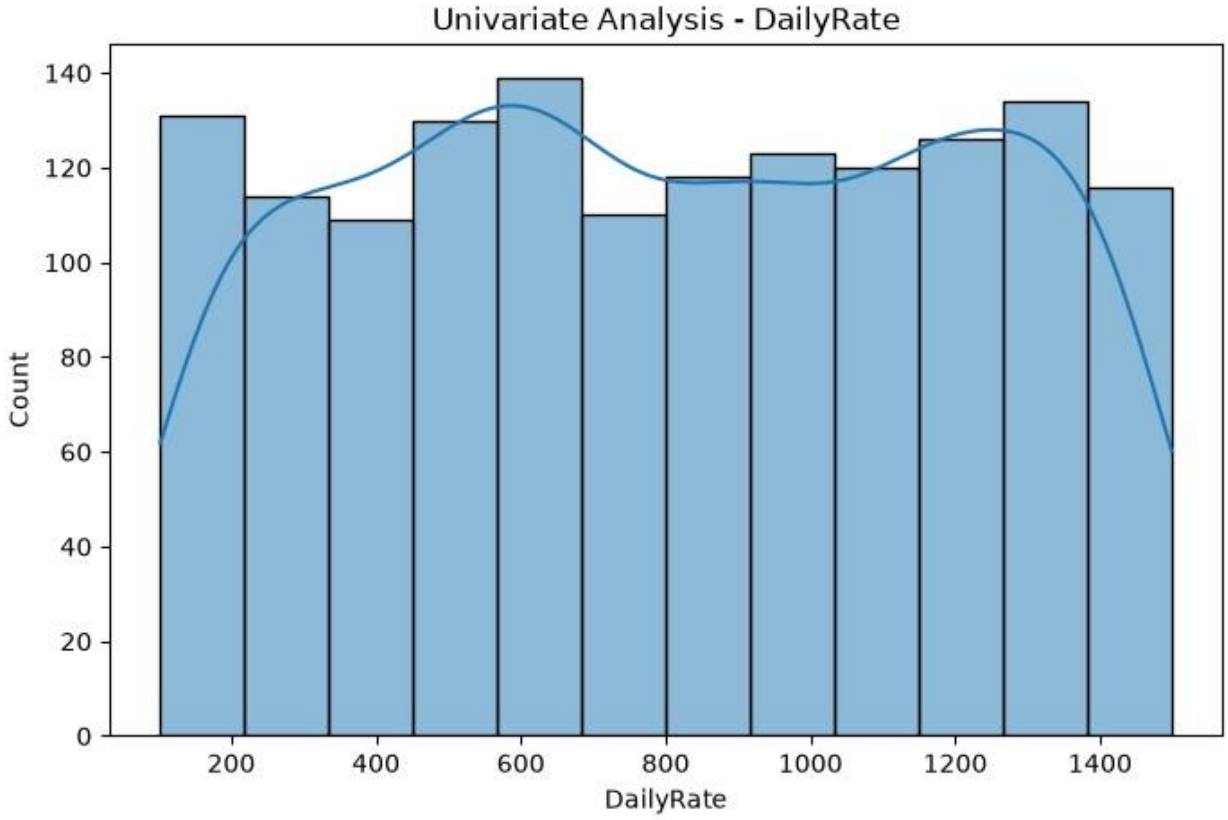
    sns.scatterplot(
data=df,
x=x,          y=y
    )

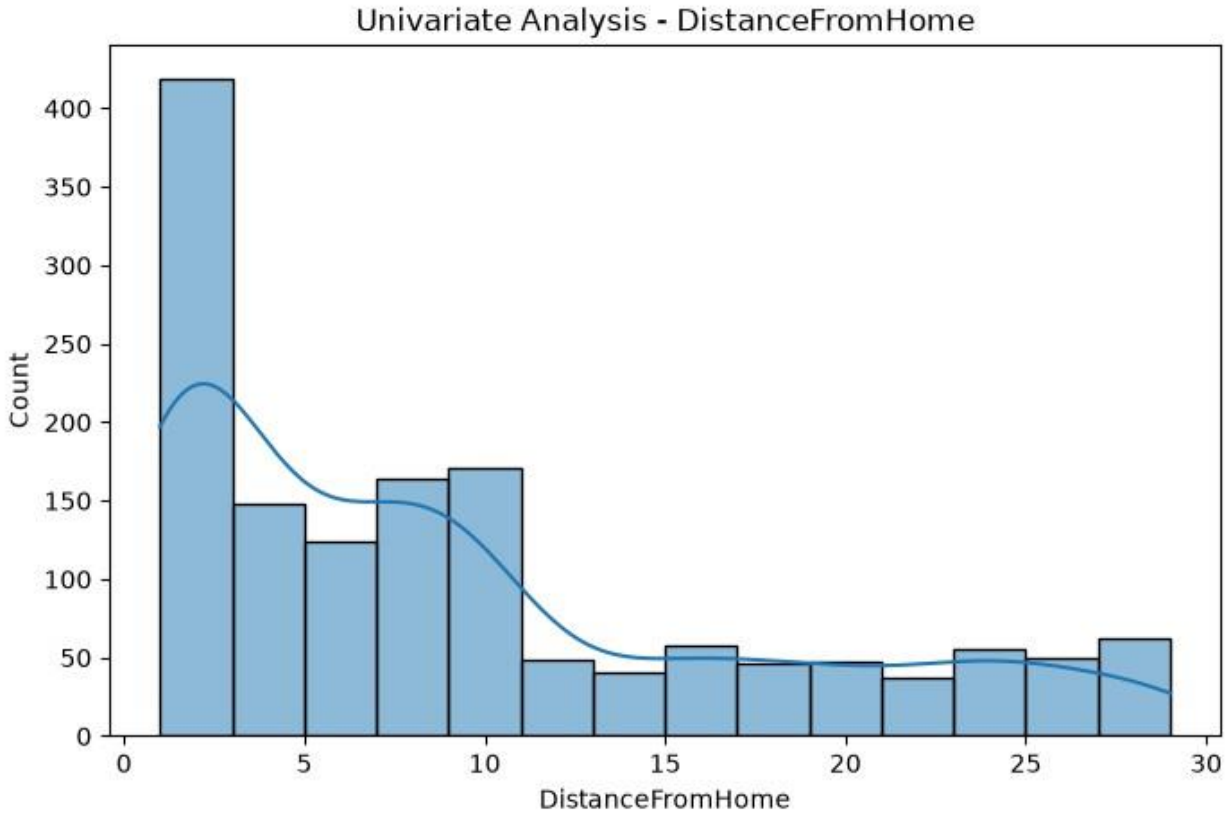
    plt.title(f"{x} vs {y}")
plt.xlabel(x)
plt.ylabel(y)     plt.show()
```

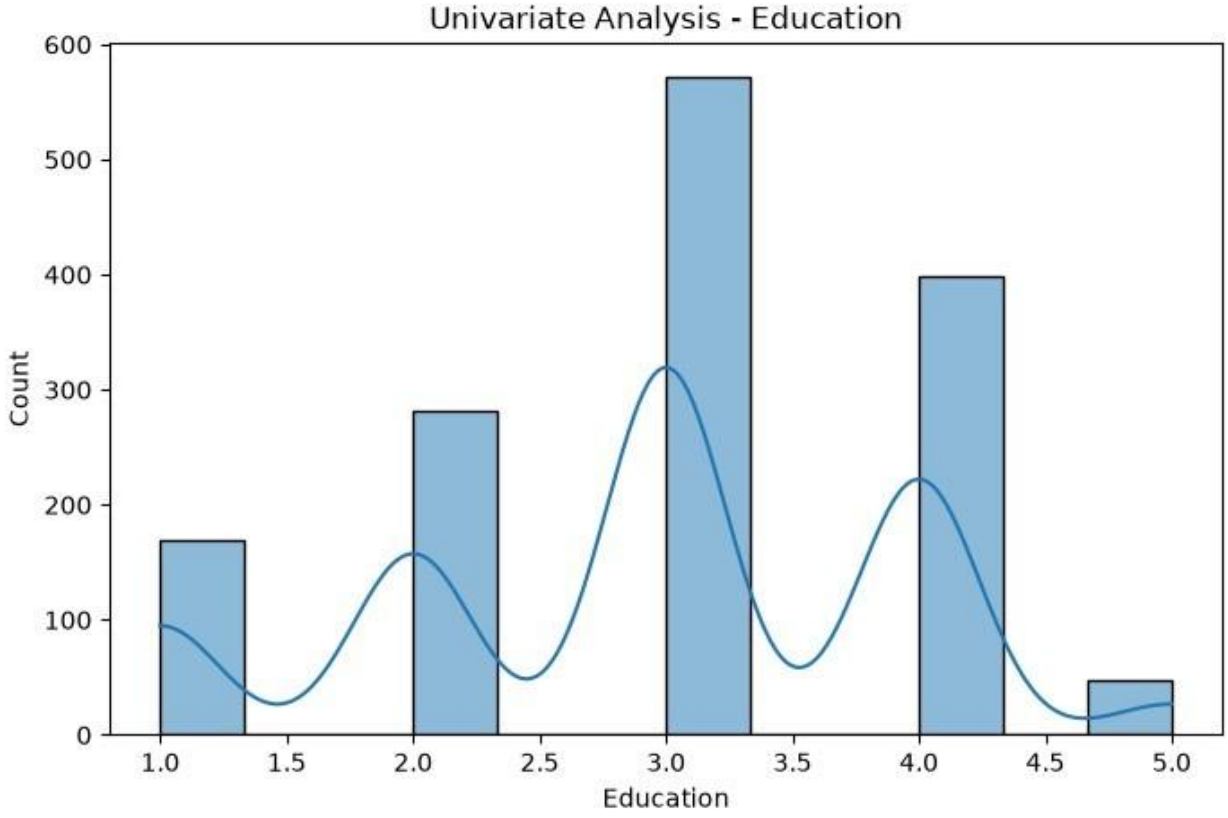


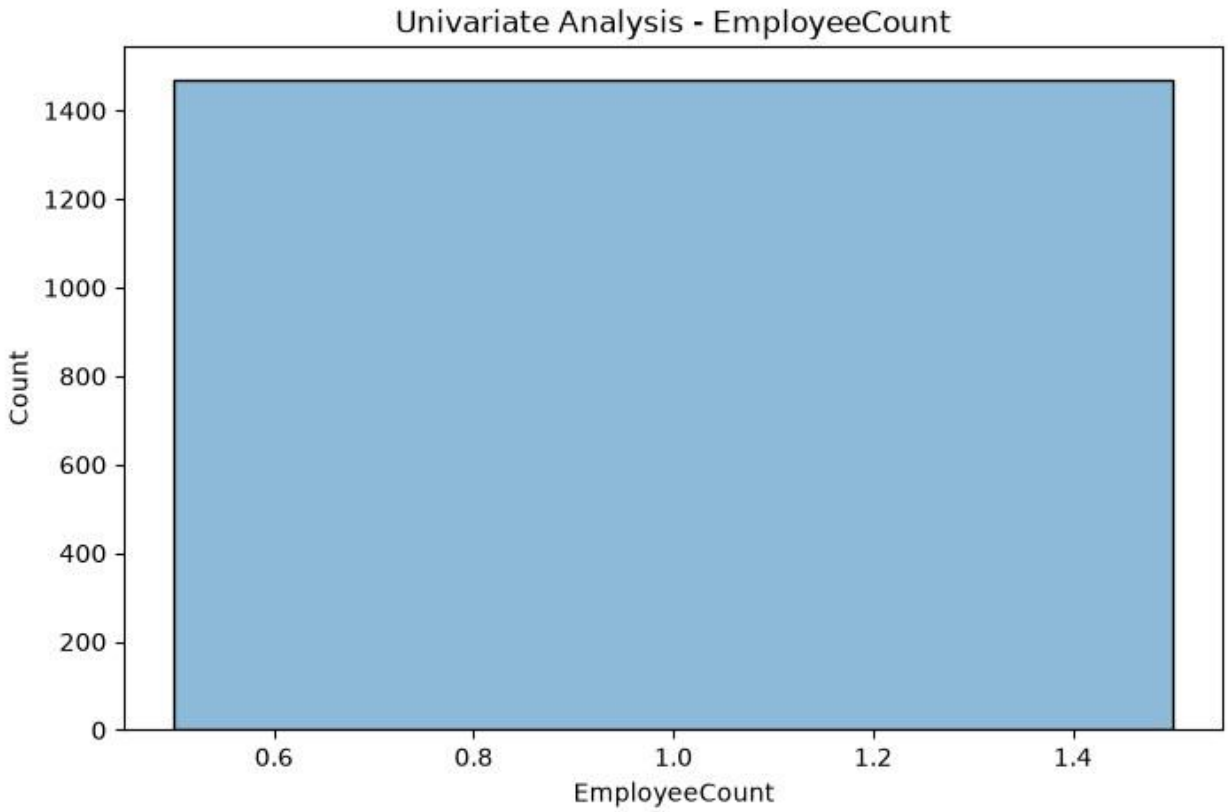

```
for column in numerical_columns:
plt.figure(figsize=(8, 5))
sns.histplot(data=df, x=column, kde=True)
plt.title(f"Univariate Analysis - {column}")
plt.show()
```

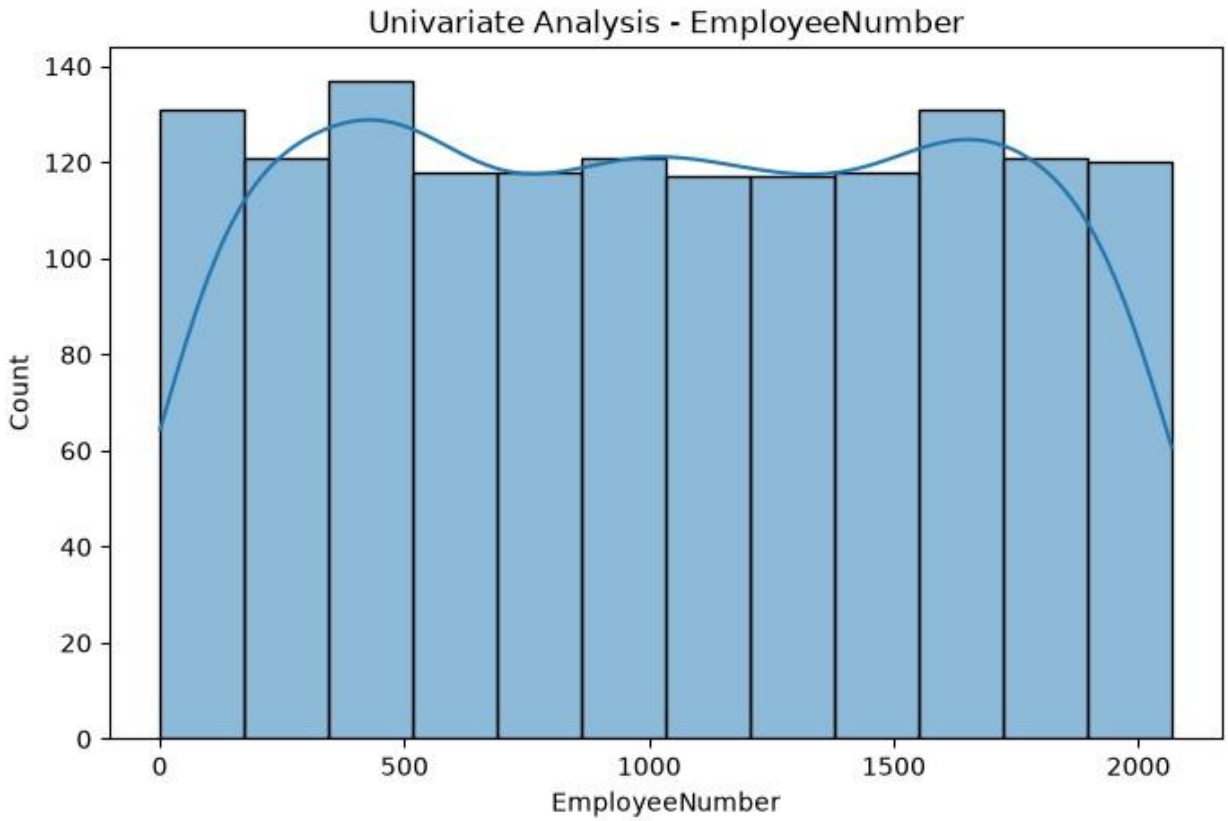

Univariate Analysis - Age
Count
Age
0
20
40
60
80
100
120
140
160
20
30
40
50
60


Univariate Analysis - DailyRate
Count
DailyRate
0
20
40
60
80
100
120
140
200
400
600
800
1000
1200
1400


Univariate Analysis - DistanceFromHome
Count
DistanceFromHome
0
50
100
150
200
250
300
350
400
0
5
10
15
20
25
30


Univariate Analysis - Education
Count
Education
0
100
200
300
400
500
600
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Univariate Analysis - EmployeeCount
Count
EmployeeCount
0
200
400
600
800
1000
1200
1400
0.6
0.8
1.0
1.2
1.4


Univariate Analysis - EmployeeNumber
Count
EmployeeNumber
0
20
40
60
80
100
120
140
0
500
1000
1500
2000

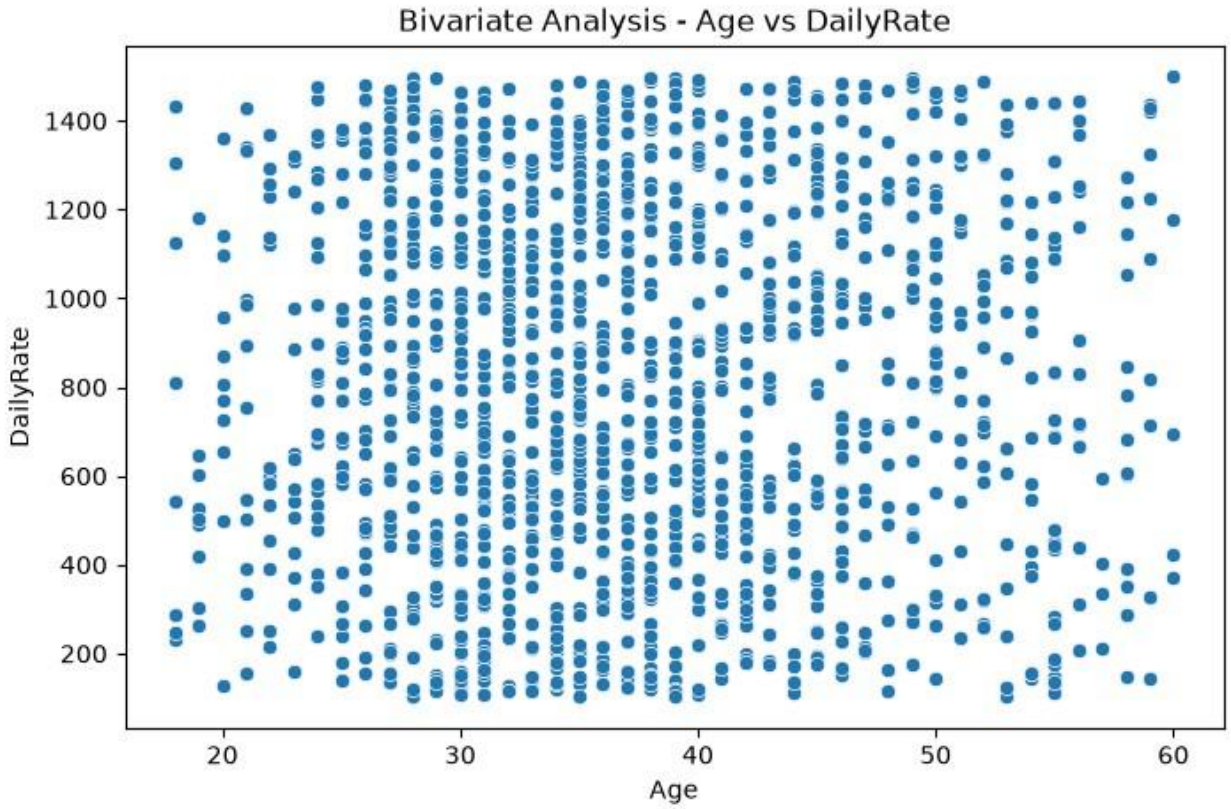
```
if len(numerical_columns) >= 2:

    x = numerical_columns[0]
y = numerical_columns[1]

    plt.figure(figsize=(8, 5))

    sns.scatterplot(
data=df,
x=x,          y=y
    )

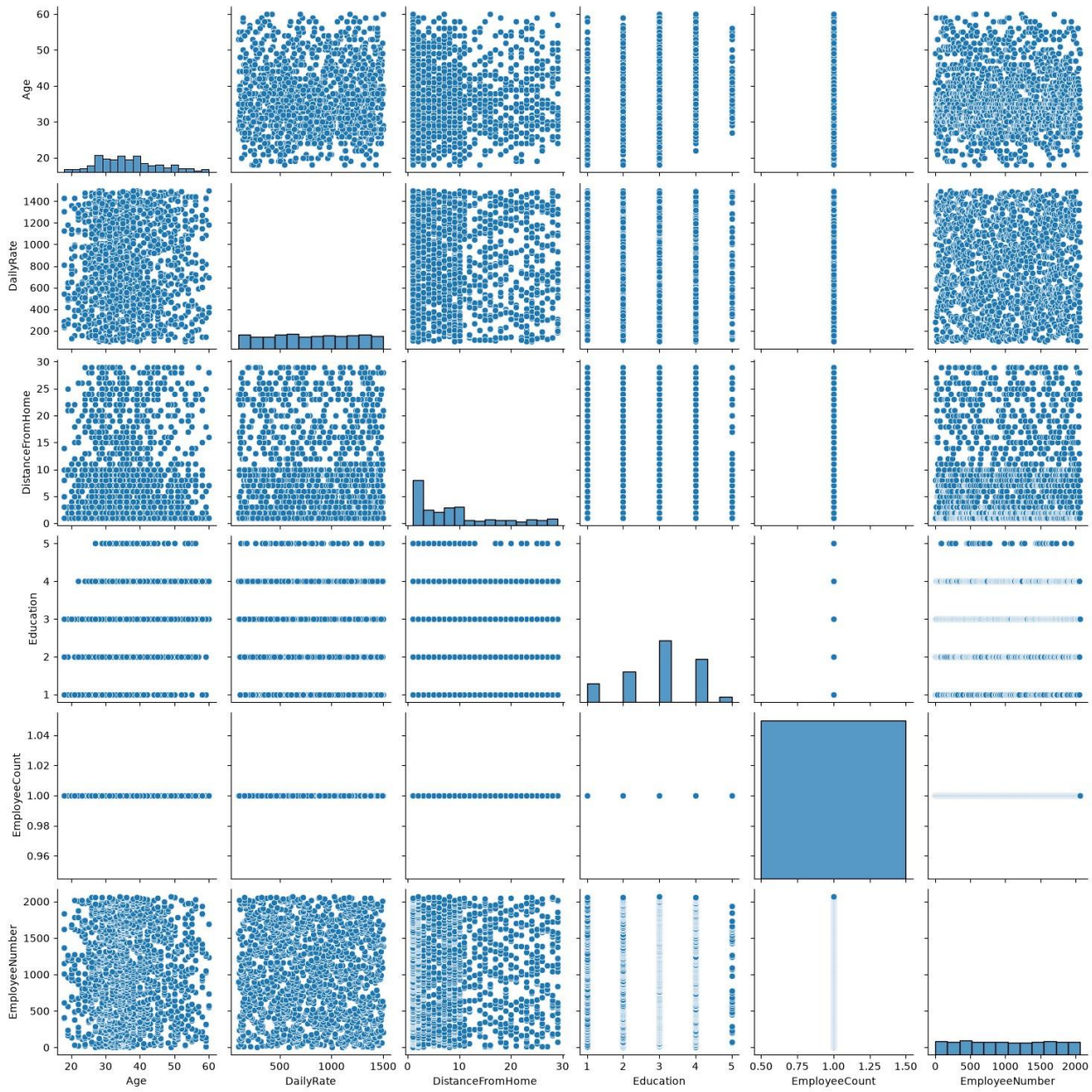
    plt.title(f"Bivariate Analysis - {x} vs {y}")
plt.show()
```




```
if len(numerical_columns) >= 3:

    sns.pairplot(
        df[numerical_columns].dropna()
    )

plt.show()
```




```
correlation_matrix = df[numerical_columns].corr()

print("Correlation with numerical variables:")
print(correlation_matrix)
```

```
Correlation with numerical variables:
                          Age  DailyRate  DistanceFromHome  Education  \
Age                  1.000000   0.010661         -0.001686   0.208034
DailyRate            0.010661   1.000000         -0.004985  -0.016806
DistanceFromHome    -0.001686  -0.004985          1.000000   0.021042
Education            0.208034  -0.016806          0.021042   1.000000
EmployeeCount             NaN        NaN               NaN        NaN
```

```
EmployeeNumber   -0.010145  -0.050990          0.032916   0.042070

                  EmployeeCount  EmployeeNumber
Age                         NaN       -0.010145
DailyRate                   NaN       -0.050990
DistanceFromHome            NaN        0.032916
Education                   NaN        0.042070
EmployeeCount               NaN             NaN  EmployeeNumber
NaN        1.000000
```

```
for column in correlation_matrix.columns:
for row in correlation_matrix.index:
                if column != row:             value =
correlation_matrix.loc[row, column]
                        if
abs(value) >= 0.7:
print(
                    f"{row} and {column}: "
f"Correlation = {value:.2f}"
                )
```

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Age                 0
Attrition           0
BusinessTravel      0
DailyRate           0
Department          0
DistanceFromHome    0
Education           0
EducationField      0
EmployeeCount       0
EmployeeNumber      0
dtype: int64
```

```
missing_percentage = (
    df.isnull().sum() / len(df)
) * 100

print("\nMissing Value Percentage:")
print(missing_percentage)
```

```
Missing Value Percentage:
Age                 0.0
Attrition           0.0
BusinessTravel      0.0
DailyRate           0.0
```

```
Department           0.0
DistanceFromHome     0.0
Education            0.0
EducationField       0.0
EmployeeCount        0.0
EmployeeNumber       0.0
dtype: float64
```

QUESTION AND ANSWER

1. What is EDA and why is it performed before machine learning?
Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset using **statistics, tables, and visualizations.**
EDA is performed to:

- Understand the data.
- Find missing values and outliers.
- Identify patterns and relationships.
- Detect errors in the dataset.
- Select useful features before building a model.

2. Univariate, Bivariate, and Multivariate Analysis

- **Univariate Analysis:** Studies **one variable** at a time.
  Example: Distribution of employee ages.
- **Bivariate Analysis:** Studies the relationship between **two variables.**
  Example: Salary vs. years of experience.
- **Multivariate Analysis:** Studies **three or more variables** together.
  Example: Salary based on age, experience, and education.

3. What insights can be obtained from a correlation heatmap?
A correlation heatmap shows the **strength and direction of relationships** between numerical variables.
It helps identify:

- Strong positive correlations.
- Strong negative correlations.
- Weak or no correlations.
- Highly related features that may contain redundant information.

4. Purpose of histograms, box plots, and scatter plots

- **Histogram:** Shows the **distribution and frequency** of numerical data.
- **Box Plot:** Shows **median, quartiles, spread, and outliers.**
- **Scatter Plot:** Shows the **relationship between two numerical variables** and helps identify patterns or outliers.

5. How can EDA identify data quality issues?
EDA helps detect:

- Missing values.
- Duplicate records.
- Outliers.
- Incorrect data types.
- Inconsistent values.
- Unusual or incorrect observations.

This allows problems to be corrected before analysis or machine learning.

6. Why is correlation important? Can it imply causation?
Correlation is important because it helps identify relationships between variables and can assist in **feature selection and prediction.** However, **correlation does not imply causation**. Two variables may be correlated without one causing the other.
Example: Ice cream sales and swimming accidents may both increase in summer, but ice cream sales do not cause swimming accidents.

7. Which visualization is suitable for categorical and numerical variables?

- **Categorical variables:** Bar charts or count plots are suitable because they show the frequency of each category.
- **Numerical variables:** Histograms and box plots are suitable because they show distribution and spread.
- **Categorical + numerical:** Box plots or violin plots can compare numerical values across categories.

8. What business insights can be derived through EDA?
From datasets such as **Netflix, Superstore, or HR Analytics**, EDA can identify:

- Best-selling products.
- Most profitable regions.
- Popular Netflix genres.
- Customer preferences.
- Employee turnover/attrition patterns.
- Sales and revenue trends.
- Factors related to customer or employee behavior.

These insights help organizations make better business decisions.

9. How does EDA contribute to feature selection and model building?
EDA helps identify **important, irrelevant, or highly correlated features**. It also detects missing values and outliers that may affect model performance.
As a result, EDA helps create a **cleaner and more useful dataset,** leading to better machine learning models.

10. Challenges of EDA on large-scale datasets

Common challenges include:

- Very large data volume.
- High processing time.
- Missing and inconsistent data.
- Too many variables.
- Difficulty in visualizing large datasets.
- High memory and computational requirements.
- Complex data from multiple sources.